Spaghetti Worms Natural Selection

Aim: To model natural selection in the environment.

Equipment:

- 20 pieces of coloured spaghetti (10 each in two colours)
- Ruler
- Scissors
- A quiet grassed area

Safety Information



Pasta – possible chemical contamination

Do not consume anything made in or for laboratory use.

Method:

Step 1: Measure ten pieces of the first colour of spaghetti and cut to an equal length.

Step 2: Measure ten pieces of the second colour of spaghetti and cut to an equal length as the first coloured spaghetti.

Step 3: Take the 20 pieces of spaghetti (with your teacher's permission) out to the chosen area on the field.

Step 4: Using a random method, throw the spaghetti into the grass. Make sure you know where it has landed (you might sketch a diagram).

Step 5: After 24 hours return and count the spaghetti. Record your observations into the table.

Step 6: Repeat set 5 another 4 times.

Often organisms of the same species have some differences. These differences are called variation. Most variation is a result of mutation in the genetic information of the organism. Sometimes the particular characteristic which is varied can be an advantage and helps the organisms adapt and survive in their environment.

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Questions

1. Describe the results you have collected.
2. Why do you think some of the 'worms' survived longer than the others?
3. What other factors could affect the survival of an organism in its environment?



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Extra Challenge

4. Make a list of the variation between the two rabbits in the photograph.
5. The fox is a predator of the rabbits. Suggest which variant is an advantage for survival in the rabbits and explain why.

Results

Colour of 'worms'	Number of 'worms'					
	0 hrs	1 day	2 days	3 days	4 days	5 days

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Answers

1. Describe the results you have collected.
   See student's results table.

2. Why do you think some of the 'worms' survived longer than the others?
   One colour will offer greater camouflage in the grass, making them less visible to predators (birds), so less likely to be eaten.

3. What other factors could affect the survival of an organism in its environment?
   Competition for food, heat, light, shelter, mates. Predators, population size of other prey.

Extra Challenge

4. Make a list of the variation between the two rabbits in the photograph.
   Colour, fur length, ear size, body size, mass.

5. The fox is a predator of the rabbits. Suggest which variant is an advantage for survival in the rabbits and explain why.
   Black colour would provide better camouflage at night time / low light when foxes hunt for rabbits.